OPTERA AI

Autonomous Multi-Agent AI Productivity Platform

Detailed Project Documentation

Project Information	Details
Project	OpteraAI
Domain	Artificial Intelligence & Machine Learning
Architecture	Multi-Agent AI Productivity Platform
Interface	Unified Dashboard
Team Name	Luminex
Team Leader	Dharani Vasan V
Team Members	• Maheswara Pandiyan A • Sivaganesh B • Santhosh A P

1. Project Overview

OpteraAI is a unified multi-agent AI productivity platform designed to connect information, personal context and digital actions through a single dashboard. Instead of treating email, documents, meetings, career activities, learning, applications, calendar events and notifications as isolated tasks, OpteraAI organizes them into coordinated workflows.

The platform uses specialized agents for specific responsibilities. Agents can process information, pass relevant context to other agents and contribute to multi-step workflows. A unified dashboard provides the main user interface so users do not need to open separate agent dashboards.

1.1 Vision

Move productivity assistance from isolated AI interactions toward context-aware, coordinated and authorized execution.

1.2 Objectives

- Provide one unified interface for major AI productivity functions.
- Reduce repetitive manual work across information and productivity workflows.
- Maintain useful personal context across documents, career, learning and other activities.
- Enable specialized agents to collaborate on multi-step tasks.
- Support authorized automation while keeping user control over consequential actions.
- Provide monitoring, scheduling, notifications and analytics.

2. Problem Statement

Modern users manage emails, documents, meetings, applications, career activities, learning and schedules across different services. Information is fragmented, and completing a workflow often requires manually reading information, transferring details between applications and tracking the next action.

- Important information can be difficult to identify among large volumes of incoming content.
- Users repeatedly transfer personal and resume information into forms and applications.
- Meeting information, documents, deadlines and follow-up actions are often disconnected.
- Personal knowledge is spread across files and services rather than being available as shared context.
- Users must switch between multiple tools and dashboards to complete related tasks.

OpteraAI addresses this gap by combining specialized agents, shared context, workflow coordination and a unified dashboard.

3. Proposed Solution

OpteraAI organizes productivity into intelligent understanding, personal intelligence, automated action and unified coordination.

3.1 Intelligent Understanding

Watcher, Classification, Priority and Research process incoming information, determine category and importance, and identify useful context or actions.

3.2 Personal Intelligence

Document, Knowledge, Enrichment, Resume, Career and Learning support document understanding, personal knowledge retrieval, profile enrichment, resume improvement, career guidance and learning.

3.3 Automated Action

Meeting, Application, Calendar and Notification convert relevant information into actions. The Meeting Agent can use authorized user context and instructions. The Application Agent can use authorized resume/profile information for supported form workflows.

3.4 Unified Coordination

Supervisor and Analytics provide coordination and activity visibility, while the unified dashboard is the main user interface.

4. Agent Ecosystem

The project materials describe OpteraAI as a 16-agent system. The final implementation should keep the approved agent count and naming consistent across code, documentation and presentation.

Agent	**Primary Responsibility**
Watcher Agent	Monitor incoming information and events
Classification Agent	Categorize incoming information
Priority Agent	Determine importance and urgency
Research Agent	Investigate and enrich relevant information
Search Agent	Search available information sources
Document Agent	Process and manage documents
Knowledge Agent	Retrieve relevant personal knowledge
Enrichment Agent	Add useful context to user/profile information
Resume Agent	Manage and improve resume information
Meeting Agent	Assist with meeting participation and context
Career Agent	Support career-related workflows
Learning Agent	Support personalized learning workflows
Application Agent	Automate supported application workflows
Calendar Agent	Manage events and scheduling
Notification Agent	Deliver reminders and alerts
Analytics Agent	Track activity and productivity
Supervisor Agent	Coordinate agent workflows

5. Core Workflows

5.1 Information-to-Action Workflow

Watcher → Classification → Priority → Research → Action Agent → Calendar/Notification → Analytics

5.2 Career and Application Workflow

Opportunity → Research → Enrichment → Resume → Application → Calendar → Notification

5.3 Knowledge Workflow

Document → Knowledge → Enrichment → Career/Learning/Research/Meeting

5.4 Meeting Workflow

Join → Listen → Understand → Retrieve Context → Answer

5.5 Scheduling Workflow

Task/Deadline → Calendar → Notification → Analytics

6. Autonomous Meeting Agent

The Meeting Agent is designed beyond simple transcription. When configured and authorized, it can participate in a meeting, process meeting content, use predefined context and instructions, and generate responses to relevant questions.

- Authorized meeting participation.
- Audio and speech processing where implemented.
- Context retrieval from permitted user information.
- Question/topic detection.
- Response generation using configured instructions and context.
- Voice or meeting response where supported.
- Post-meeting notes and action extraction where implemented.

Meeting participation must respect platform permissions, meeting policies and user authorization.

7. Application Automation

The Application Agent is intended to reduce repetitive form-filling work. It can use structured information from the user's profile and resume to identify relevant form fields and populate supported applications.

- Identify a supported application workflow.
- Extract relevant authorized profile and resume information.
- Map information to application fields.
- Fill the supported form.
- Validate important fields.
- Request or enforce user authorization for consequential submission.
- Record application status and follow-up information.

Fully autonomous submission should only be enabled where the platform, website and user permissions allow it.

8. Personal Knowledge and RAG

OpteraAI can use a personal knowledge layer to make relevant user-provided information available to agents. A typical implementation can ingest documents, split content, create embeddings, store them in a vector index and retrieve relevant context.

- Document ingestion
- Text extraction and preprocessing

- Chunking
- Embedding generation
- Vector storage
- Similarity retrieval
- Context assembly
- Agent response or action

9. Unified Dashboard

The unified dashboard is the primary interface for OpteraAI. Users should not be redirected to separate agent dashboards for normal operations.

- Authentication and account access.
- Unified sidebar navigation.
- Email and information monitoring.
- Documents and personal knowledge.
- Career, resume and learning functions.
- Meeting controls and meeting context.
- Applications and application status.
- Calendar and deadlines.
- Notifications and preferences.
- Analytics and activity.
- Agent/workflow status.
- Logout and account controls.

10. System Architecture

A high-level architecture consists of the unified frontend, backend/API layer, agent services, shared data and context layer, external integrations and AI/model services.

Layer	Role
Unified Frontend	Single interface for all platform capabilities
API / Gateway	Authentication, routing and backend communication
Agent Layer	Specialized domain responsibilities
Coordination Layer	Supervisor/workflow routing and handoff
Data Layer	Users, documents, tasks, applications, events and notifications
Knowledge Layer	Personal knowledge retrieval and context
External Integrations	Email, calendar, OAuth and browser automation
AI Services	LLM, embeddings, speech and other model capabilities

11. Technology Stack

Area	Technology / Tool
Frontend	React, Vite, TypeScript, Tailwind CSS

Backend	Python, FastAPI, SQLAlchemy
AI Models	Groq, Gemini
Embeddings / RAG	Sentence Transformers, ChromaDB
Database	PostgreSQL
Browser Automation	Playwright
Speech / Audio	Whisper, FFmpeg
Authentication	Google OAuth, JWT
Email	Gmail API
Scheduling	APScheduler

12. Security and User Control

- Authenticated access and user-level data isolation.
- OAuth for supported external services.
- API keys and credentials stored in environment configuration.
- Authorization for consequential external actions.
- Controls for automation, meeting participation and notifications.
- Activity logging for important workflows.
- Careful handling of sensitive personal information.

13. Data Management

- User and authentication data
- Email/message metadata
- Document metadata and content references
- Knowledge and retrieval records
- Resume/profile information
- Career opportunities and applications
- Meeting records and action items
- Calendar events and deadlines
- Notification preferences and delivery records
- Agent execution logs
- Analytics and productivity metrics

14. Agent-to-Agent Communication

Agents collaborate through structured task handoffs and shared context. This reduces the need for users to manually copy information between separate dashboards.

- Watcher identifies new information.
- Classification categorizes it.
- Priority determines importance.
- Research adds context.
- A domain agent performs the required action.
- Calendar and Notification handle follow-up.
- Analytics records outcomes.
- Supervisor coordinates routing and status.

15. Existing Solutions and Differentiation

Platform	Primary Strength	OpteraAI Differentiation
Microsoft 365 Copilot	Email, documents, meetings and productivity	Broader multi-agent personal workflow spanning productivity, knowledge, career, applications and learning.
Notion AI	Knowledge, documents and workspace assistance	Connects knowledge with downstream career, application, meeting and productivity actions.
Otter.ai	Meeting transcription and intelligence	Connects meeting context to a wider personal productivity workflow.
Teal	Resume and career management	Connects resume/career context with research, applications, calendar and notifications.
Motion	Task, calendar and scheduling	Combines scheduling with information, knowledge, career and application workflows.

The intended differentiator is the connection of multiple productivity domains through shared context and a unified workflow, rather than simply the number of agents.

16. Key Advantages

- Unified dashboard instead of separate agent interfaces.
- Specialized agents for domain-specific tasks.
- Shared personal context across relevant workflows.
- Agent-to-agent coordination for multi-step tasks.
- Application automation for supported workflows.
- Context-aware meeting assistance.
- Integrated calendar and notification management.
- Personalized career and learning support.
- Analytics for workflow outcomes.

17. Use Cases

17.1 Student / Learner

Manage learning goals, documents, opportunities, meetings, deadlines and notifications from one platform.

17.2 Job Seeker

Research opportunities, maintain resume context, prepare applications and track deadlines and follow-ups.

17.3 Working Professional

Process work information, prepare for meetings, manage schedules and track follow-up actions.

17.4 Knowledge Worker

Retrieve personal knowledge from documents and connect research findings with ongoing productivity tasks.

18. Limitations and Considerations

- External APIs can impose rate limits or availability constraints.
- Browser-based application automation depends on website structure and permissions.
- AI-generated responses may require validation for high-impact actions.
- Meeting participation depends on technical and authorization constraints.
- Sensitive personal information requires careful storage and access control.
- Agent coordination requires robust error handling and observability.

19. Project Roadmap

Phase	Focus
Phase 1	Problem definition, research and architecture
Phase 2	Core information-processing agents
Phase 3	Knowledge, personalization and career agents
Phase 4	Meeting and productivity automation
Phase 5	Unified dashboard integration and production testing
Phase 6	Scalability, integrations and future capabilities

20. Future Scope

- Mobile and cross-platform clients.
- Additional email, calendar, communication and career integrations.
- Advanced long-term personal memory.
- Multilingual interaction.
- Improved voice-based workflows.
- More autonomous workflows with granular user controls.
- Advanced productivity analytics and proactive recommendations.
- Scalable deployment for larger user populations.

21. Conclusion

OpteraAI proposes a unified approach to AI-assisted productivity in which specialized agents collaborate around a shared understanding of the user's information and goals. The platform connects information processing, personal knowledge, career and learning support, meeting assistance, application automation, scheduling and notifications through one dashboard.

The core idea is to move beyond isolated AI assistance toward coordinated workflows in which information can be understood, enriched and converted into authorized actions.